

3. Perform basic Image Handling and processing operations on the image • Read an image in Python and Convert an Image to show outline using Canny function.

Aim:

To perform basic image handling operations by reading an image in Python and detecting its outlines (edges) using the Canny Edge Detection algorithm in OpenCV.

Procedure:

- Import the required Python libraries (OpenCV and Matplotlib).
- Read the input image using the cv2.imread() function and convert it to grayscale.
- Apply the cv2.Canny() function with suitable threshold values to detect the edges.
- Display both the original image and the detected edge image.
- Observe that the important outlines and boundaries of objects are highlighted in the output image.

Code:

```
import cv2

import matplotlib.pyplot as plt

# Read the image

image = cv2.imread("/Users/michellejoanna/Downloads/meee.jpeg")

# Convert the image from BGR to RGB

rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Convert the image to Grayscale

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Apply Canny Edge Detection

edges = cv2.Canny(gray, 100, 200)
```

```
# Display the original and edge-detected images
```

```
plt.figure(figsize=(10, 5))
```

```
plt.subplot(1, 2, 1)
```

```
plt.imshow(rgb)
```

```
plt.title("Original Image")
```

```
plt.axis("off")
```

```
plt.subplot(1, 2, 2)
```

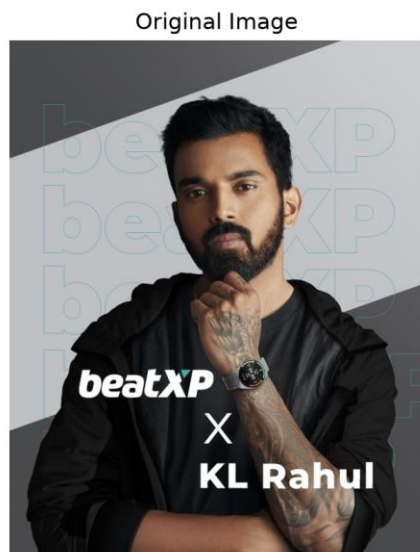
```
plt.imshow(edges, cmap="gray")
```

```
plt.title("Canny Edge Detection")
```

```
plt.axis("off")
```

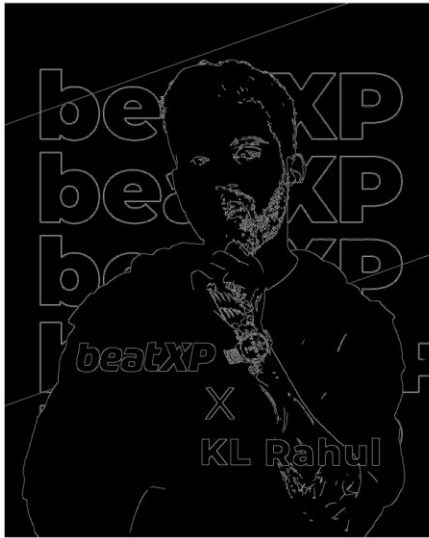
```
plt.show()
```

Sample Input:



Sample Output:

Canny Edge Detection



Result:

The input image was successfully read using Python, converted to grayscale, and processed using the Canny Edge Detection algorithm.